TEXORA · COURSE NOTES · LECTURE SHEET

Chemical Kinetics

How fast reactions go and why — rate laws and orders, integrated equations and half-lives, activation energy, and the Arrhenius law.

CHE-1105 · CHEMISTRY-I · TOPIC 7 · LEVEL 1 TERM 1

1. Rate and the rate law

Equilibrium (Topic 6) says where a reaction is heading; kinetics says **how fast** it gets there — and the two are independent: diamond is thermodynamically unstable relative to graphite, yet persists for ages because the rate is negligible. The **rate** of a reaction is the change of concentration per unit time. Experiment finds, for very many reactions, a **rate law** of the form

$$\text{rate} = k\,[A]^m\,[B]^n$$

where k is the **rate constant** (fixed at a given temperature), and m, n are the **orders** with respect to A and B; m + n is the overall order. Orders are experimental quantities — they are *not* in general the stoichiometric coefficients. **Molecularity**, by contrast, is a theoretical count of the molecules taking part in a single elementary step (unimolecular, bimolecular, termolecular); for an elementary step, order and molecularity coincide, and complex reactions are chains of such steps whose slowest — the **rate-determining step** — sets the overall rate.

WORKED EXAMPLE — ORDER FROM INITIAL RATES

Initial rates for A + B → products: (1) [A] = 0.10, [B] = 0.10 → rate 0.020; (2) [A] = 0.20, [B] = 0.10 → rate 0.080; (3) [A] = 0.10, [B] = 0.20 → rate 0.040 (mol L^{-1} s^{-1} and mol L^{-1}). Doubling [A] alone multiplies the rate by 4 → second order in A. Doubling [B] alone doubles the rate → first order in B. Rate law: rate = k$[A]^2[B]$, overall order 3, and from experiment 1, k = 0.020 / ($0.10^2 \times 0.10$) = **20 L^2 mol^{-2} s^{-1}**. The method of initial rates is the standard experimental route to every rate law.

2. Integrated rate laws and half-life

Integrating the rate law gives concentration as a function of time — and each order has its own signature:

ORDER	INTEGRATED LAW	STRAIGHT-LINE PLOT	HALF-LIFE $T_{½}$
Zero	$[A] = [A]_0 - kt$	[A] vs t	$[A]_0 / 2k$ — falls as reaction proceeds
First	$\ln [A] = \ln [A]_0 - kt$	ln [A] vs t	$\ln 2 / k \approx 0.693/k$ — constant, independent of $[A]_0$
Second	$1/[A] = 1/[A]_0 + kt$	1/[A] vs t	$1 / (k[A]_0)$ — rises as reaction proceeds

The **half-life** — the time for half the reactant to be consumed — is the most quoted kinetic number in science. For a first-order reaction its constancy is diagnostic: equal successive half-lives prove first-order behaviour whatever the starting concentration, which is why radioactive decay (and first-order dye hydrolysis, and drug elimination) is tracked in half-lives. A reaction with $t_{½}$ = 20 min has $k = 0.693/20 \approx 0.0347\ \text{min}^{-1}$; after three half-lives (60 min) exactly $(1/2)^3 = 1/8$ of the reactant survives.

3. Temperature: collision theory and Arrhenius

Reactions speed up with temperature far more than collision frequency alone explains, because molecules must collide with energy above a threshold — the **activation energy** E_a — and in a favourable orientation. Arrhenius compressed this into one equation:

$$k = A \cdot e^{-E_a/RT}$$

where A (the frequency factor) counts properly oriented collisions and the exponential counts the fraction energetic enough to react. Two consequences dominate practice: even a modest E_a makes rates roughly **double for a 10 K rise** near room temperature (the dyer's rule of thumb for why hot baths run so much faster); and taking logarithms, ln k plotted against 1/T gives a straight line of slope $-E_a/R$ — the standard measurement of activation energy. In **transition-state** language, reactants climb to an activated complex at the energy summit and fall to products on the other side; E_a is the height of the pass, for the forward and reverse reactions differing by ΔH.

4. Catalysis in kinetic terms

A catalyst works by opening a **different path with a lower E_a**; from the Arrhenius equation, lowering E_a raises k exponentially, while ΔH and the equilibrium position (Topic 6) stay untouched. Catalysis divides into **homogeneous** (catalyst in the same phase — acid catalysis of ester hydrolysis) and

heterogeneous (a surface — iron in the Haber process, platinum in catalytic converters), with enzymes as biology's extreme case: enormous rate enhancements at body temperature with total specificity. Inhibitors do the reverse, and every stored chemical — from dye to monomer — ships with kinetics in mind.

5. The textile thread

Dyeing is kinetics wearing overalls: the **strike** (initial uptake rate) decides levelness, so dyers control it with temperature ramps — often 1 °C per minute — and with levelling agents that temporarily slow the fastest sites; reactive-dye fixation and hydrolysis are two competing rates whose ratio the alkali dose sets; and the fading of a fabric in sunlight, the tendering of cotton by acid, and the shelf life of every finish are first-order decays with half-lives the engineer can measure and extend. Kinetics is where this course meets the stopwatch of the production floor.

6. Glossary — key terms of this topic

TERM	MEANING
Rate law	The experimentally found relation rate = $k[A]^m[B]^n$.
Order	The power of a concentration in the rate law; experimentally determined, not read from the equation.
Molecularity	The number of molecules in an elementary step (1, 2 or 3).
Rate-determining step	The slowest step of a mechanism, which governs the overall rate.
Half-life	The time for half of a reactant to be consumed; constant for first-order reactions.
Activation energy	The minimum collision energy for reaction; the barrier height in the Arrhenius equation.
Activated complex	The fleeting highest-energy arrangement at the top of the reaction path.

7. Key points and summary

- Rate laws and orders are experimental (initial-rates method); molecularity belongs to single steps, and the slowest step rules.
- Each order has its integrated law and half-life signature; first-order half-life is concentration-independent — the great diagnostic.
- Temperature acts through the Arrhenius exponential: rates roughly double per 10 K, ln k vs 1/T measures E_a, and catalysts work by lowering E_a along a new path without touching equilibrium.

8. Practice

Answer in your own words; brief model answers follow.

1. A reaction is second order in A and zero order in B. Write its rate law and its overall order.
2. A first-order reaction has k = 0.0693 min^{-1}. Find its half-life, and the fraction remaining after 30 minutes.
3. Successive measurements of a decaying reactant show half-lives of 12 min, 12 min, 12 min. What order is indicated, and why?
4. For a second-order reaction (single reactant), write the integrated law and state what plot gives a straight line.
5. Using the Arrhenius idea, explain why a 10 °C hotter dyebath can nearly double the dyeing rate — and one risk this creates for levelness.
6. Distinguish order from molecularity with the example of a reaction that is overall second order but proceeds in two steps.
7. A catalyst is added to an equilibrium mixture. State its effect on (a) time to reach equilibrium, (b) equilibrium composition, (c) E_a of the reverse reaction.
8. From the worked example of §1, predict the initial rate when [A] = 0.30 and [B] = 0.10 mol L^{-1}.

Answer key

1. rate = $k[A]^2$; overall order 2 (zero order in B contributes nothing).
2. $t_{½}$ = 0.693 / 0.0693 = 10 min. 30 min = 3 half-lives → $(1/2)^3$ = 1/8 remains.
3. First order — only first-order reactions have a concentration-independent (hence constant) half-life.
4. $1/[A] = 1/[A]_0 + kt$; a plot of 1/[A] against t is linear (slope k).
5. The fraction of collisions energetic enough to react grows roughly exponentially with T (Arrhenius), doubling k for a typical E_a over 10 K. Risk: a fast strike — dye rushes onto the first fibre it meets, giving unlevel shades; hence controlled temperature ramps.
6. Order is the measured concentration dependence of the whole reaction; molecularity counts molecules in one elementary step. A two-step reaction can be overall second order (e.g. rate set by a bimolecular slow step) while including a unimolecular step as well.
7. (a) Shortened. (b) Unchanged. (c) Lowered equally — the catalyst lowers the barrier in both directions.
8. rate = $20 \times (0.30)^2 \times 0.10 = 20 \times 0.009 = 0.18$ mol L^{-1} s^{-1}.